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Bewijs dat $\text{Bgtan } \frac{1}{3} = \text{Bgtan } \frac{1}{5} + \text{Bgtan } \frac{1}{8}$

$$\text{Rechterzijde} = \text{Bgtan} \left(\frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \cdot \frac{1}{8}} \right)$$

$$= \text{Bgtan} \left(\frac{\frac{13}{40}}{1 - \frac{1}{40}} \right)$$

$$= \text{Bgtan} \left(\frac{13}{40} \cdot \frac{40}{39} \right)$$

$$= \text{Bgtan} \left(\frac{13}{39} \right) = \text{Bgtan} \left(\frac{1}{3} \right)$$

$$\text{Linkerzijde} = \text{Bgtan} \left(\frac{1}{3} \right)$$

$$\Rightarrow \text{L.L} = \text{R.L}$$

Q.E.D.

References